

**PAPER\_722: UQFF Quantum Variable Documents: M\_bh, mu\_j, P\_core, t\_n, pi**

**Class:** UQFFKnowledgeBaseKB8 **CP4 Entry:** #306 **Keywords:** UQFF, quantum variables, black hole mass, magnetic moment, P\_core, normalized time, pi quantum encoding, SMBH **Session:** 176 | **Version:** v5.33 **Source:** grok\_share\_ba508f76c8e.txt

**Abstract**

Five quantum variable document pages define:  $M_{bh}$  (Milky Way SMBH mass for U\_g4 scaling),  $\mu_j$  (magnetic string dipole moment),  $P_{core}$  (core pressure normalization for U\_g3),  $t_n$  (normalized phase time modulating all UQFF terms via  $\cos(\pi t_n)$ ), and  $\pi$  (the mathematical constant encoding 26 quantum states through  $[SSq]^{n_{26}}e^{-\pi-t}$ ).

**1. Variable Registry**

| Variable   | Value                                                          | Physical Role                                        |
|------------|----------------------------------------------------------------|------------------------------------------------------|
| $M_{bh}$   | $1.989 \times 10^{36} \text{ kg}$<br>( $\sim 10^6 M_{\odot}$ ) | Scales U_g4 AGN/Final Parsec                         |
| $\mu_j$    | $3.38 \times 10^{23} \text{ J}\cdot\text{m}$                   | Magnetic string coupling in U_m                      |
| $P_{core}$ | 1.0 (normalized)                                               | Core pressure factor in U_g3                         |
| $t_n$      | 0.0 s                                                          | Phase modulator: $\cos(\pi t_n)$                     |
| $\pi$      | 3.14159...                                                     | $[SSq]^{n_{26}} \cdot e^{-\pi-t}$ , 26-state encoder |

**2. U\_g4 Black Hole Scaling**

$$U_{g4}(t) = k_4 \cdot \frac{\rho_{SCm} \cdot M_{bh}}{d_g} \cdot e^{-\alpha t} \cdot \cos(\pi t_n) \cdot (1 + f_{feedback})$$

**3. U\_g3 Star Formation**

$$U_{g3} = k_3 \sum_j B_j(r, \theta, t, \rho_{SCm}) \cdot \cos(\omega_s t \pi) \cdot P_{core} \cdot E_{react}$$

**4. Pi as Quantum Encoder**

$$[SSq]^{n_{26}} \cdot e^{-\pi-t} \quad \text{encodes 26 quantum states via } \pi\text{-barrier}$$

**References**

- UQFF KB grok\_share\_ba508f76c8e.txt entry #72
- Session 176, v5.33

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176\*